

Research Notes on Time series forecasting

Time series forecasting

>> Section 1

Additionally, time series analysis techniques may be divided into parametric and non-parametric methods. The parametric approaches assume that the underlying stationary stochastic process has a certain structure which can be described using a small number of parameters (for example, using an autoregressive or moving-average model). In these approaches, the task is to estimate the parameters of the model that describes the stochastic process. By contrast, non-parametric approaches explicitly estimate the covariance or the spectrum of the process without assuming that the process has any particular structure. Methods of time series analysis may also be divided into linear and non-linear, and univariate and multivariate.

>> Section 2

Methods for analysis Methods for time series analysis may be divided into two classes: frequency-domain methods and time-domain methods. The former include spectral analysis and wavelet analysis; the latter include auto-correlation and cross-correlation analysis. In the time domain, correlation and analysis can be made in a filter-like manner using scaled correlation, thereby mitigating the need to operate in the frequency domain.

>> Section 3

Motivation In the context of statistics, econometrics, quantitative finance, seismology, meteorology, and geophysics the primary goal of time series analysis is forecasting. In the context of signal processing, control engineering and communication engineering it is used for signal detection. Other applications are in data mining, pattern recognition and machine learning, where time series analysis can be used for clustering, classification, query by content, anomaly detection as well as forecasting.

>> Section 4

Curve fitting is the process of constructing a curve, or mathematical function, that has the best fit to a series of data points, possibly subject to constraints. Curve fitting can involve either interpolation, where an exact fit to the data is required, or smoothing, in which a "smooth" function is constructed that approximately fits the data. A related topic is regression analysis, which focuses more on questions of statistical inference such as how much uncertainty is present in a curve that is fit to data observed with random errors. Fitted curves can be used as an aid for data visualization, to infer values of a function where no data are available, and to summarize the relationships among two or more variables. Extrapolation refers to the use of a fitted curve beyond the range of the observed

data, and is subject to a degree of uncertainty since it may reflect the method used to construct the curve as much as it reflects the observed data.

>> Section 5

Autocorrelation analysis to examine serial dependence Spectral analysis to examine cyclic behavior which need not be related to seasonality. For example, sunspot activity varies over 11 year cycles. Other common examples include celestial phenomena, weather patterns, neural activity, commodity prices, and economic activity. Separation into components representing trend, seasonality, slow and fast variation, and cyclical irregularity: see trend estimation and decomposition of time series

>> Section 6

Subsequence time series clustering Subsequence time series clustering resulted in unstable (random) clusters induced by the feature extraction using chunking with sliding windows. It was found that the cluster centers (the average of the time series in a cluster - also a time series) follow an arbitrarily shifted sine pattern (regardless of the dataset, even on realizations of a random walk). This means that the found cluster centers are non-descriptive for the dataset because the cluster centers are always nonrepresentative sine waves.

>> Section 7

A simple way to examine a regular time series is manually with a line chart. The datagraphic shows tuberculosis deaths in the United States, along with the yearly change and the percentage change from year to year. The total number of deaths declined in every year until the mid-1980s, after which there were occasional increases, often proportionately - but not absolutely - quite large. A study of corporate data analysts found two challenges to exploratory time series analysis: discovering the shape of interesting patterns, and finding an explanation for these patterns. Visual tools that represent time series data as heat map matrices can help overcome these challenges.

>> Section 8

Panel data A time series is one type of panel data. Panel data is the general class, a multidimensional data set, whereas a time series data set is a one-dimensional panel (as is a cross-sectional dataset). A data set may exhibit characteristics of both panel data and time series data. One way to tell is to ask what makes one data record unique from the other records. If the answer is the time data field, then this is a time series data set candidate. If determining a unique record requires a time data field and an additional identifier which is unrelated to time (e.g.

Research Notes on Time series forecasting

Time series forecasting

student ID, stock symbol, country code), then it is panel data candidate. If the differentiation lies on the non-time identifier, then the data set is a cross-sectional data set candidate.

>> Section 9

In mathematics, a time series is a sequence of data points indexed, listed, or graphed in chronological order. Most commonly, a time series consists of observations recorded at successive equally spaced points in time. Thus, it represents a form of discrete-time data. A time series may describe measurements collected over seconds, days, years, or even centuries. Common examples include heights of ocean tides, counts of sunspots, daily temperature readings, and the closing values of stock market indices such as the Dow Jones Industrial Average. A time series is often visualized using a run chart (a type of temporal line chart), which helps identify patterns such as trends, seasonal effects, and irregular fluctuations. Time series are widely used in statistics, actuarial science, signal processing, pattern recognition, econometrics, mathematical finance, weather forecasting, earthquake prediction, electroencephalography, control engineering, astronomy, communications engineering, and many other areas of applied science and engineering that involve temporal measurements. Time series analysis comprises methods for analyzing time series data in order to extract meaningful statistics and other characteristics of the data. Time series forecasting is the use of a model to predict future values based on previously observed values. Generally, time series data is modeled as a stochastic process. While regression analysis is often employed in such a way as to test relationships between one or more different time series, this type of analysis is not usually called "time series analysis", which refers in particular to relationships between different points in time within a single series. Time series data have a natural temporal ordering. This makes time series analysis distinct from cross-sectional studies, in which there is no natural ordering of the observations (e.g. explaining people's wages by reference to their respective education levels, where the individuals' data could be entered in any order). Time series analysis is also distinct from spatial data analysis where the observations typically relate to geographical locations (e.g. accounting for house prices by the location as well as the intrinsic characteristics of the houses). A stochastic model for a time series will generally reflect the fact that observations close together in time will be more closely related than observations further apart. In addition, time series models will often make use of the natural one-way ordering of time so that values for a given period will be expressed as deriving in some way from past values, rather than from future values (see time

reversibility). Time series analysis can be applied to real-valued, continuous data, discrete numeric data, or discrete symbolic data (i.e. sequences of characters, such as letters and words in the English language).